Exercise-1

✎ **Marked questions are recommended for Revision.**

PART - I : SUBJECTIVE QUESTIONS

Section (A) : Algebra of Complex Numbers and Its Representation and Demoivre's Theorem

A-1. Find the real values of x and y for which the following equation is satisfied :

(i) $\dfrac{(1+i)\,x-2i}{3+i}+\dfrac{(2-3i)\,y+i}{3-i}=i$

(ii) $\dfrac{x}{1+2i}+\dfrac{y}{3+2i}=\dfrac{5+6i}{8i-1}$

(iii) $(2 + 3i)\, x^2 - (3 - 2i)\, y = 2x - 3y + 5i$

(iv) $4x^2 + 3xy + (2xy - 3x^2)\, i = 4y^2 - (x^2/2) + (3xy - 2y^2)\, i$

A-2_. Let $z = \dfrac{1+2(\sin\theta)i}{1-(\sin\theta)i}$

(i) Find the number of values of $\theta\in[0, 4\pi]$ such that z is purely imaginary.

(ii) Find the sum of all values of $\theta\in[0, 4\pi]$ such that z is purely real.

A-3. (i) Find the real values of x and y for which $z_1 = 9y^2 - 4 - 10ix$ and $z_2 = 8y^2 - 20i$ are conjugate complex of each other.

(ii) Find the value of $x^4 - x^3 + x^2 + 3x - 5$ if $x = 2 + 3i$

A-4. Find

(i) the square root of 7 + 24 i (ii) $\sqrt{i}+\sqrt{-i}$

A-5. Solve the following for z : $z^2 - (3 - 2i)\, z = (5i - 5)$

A-6. Simplify and express the result in the form of a + bi :

(i) $-\, i\, (9 + 6\, i)\, (2 - i)^{-1}$ (ii) $\left(\dfrac{4i^3-i}{2i+1}\right)^2$

(iii) $\dfrac{1}{(1-\cos\theta)+2\,i\,\sin\theta}$ (iv) $\left(\sqrt{3}+i\right)e^{-i\frac{\pi}{6}}$

A-7. Convert the following complex numbers in Eulers form

(i) $z = -\pi$ (ii) $z = 5i$ (iii) $z = -\sqrt{3} - i$ (iv) $z = -2\left(\cos\dfrac{\pi}{5}+i\sin\dfrac{\pi}{5}\right)$

A-8. Find the modulus, argument and the principal argument of the complex numbers.

(i) $z = 1 + \cos\dfrac{18\pi}{25} + i\sin\dfrac{18\pi}{25}$ (ii) $z = -2\,(\cos 30° + i\sin 30°)$

(iii) $(\tan 1 - i)^2$ (iv) $\dfrac{i-1}{i\left(1-\cos\dfrac{2\pi}{5}\right)+\sin\dfrac{2\pi}{5}}$

A-9.✎ Dividing polynomial f(z) by z – i, we get the remainder i and dividing it by z + i, we get the remainder 1 + i. Find the remainder upon the division of f(z) by $z^2 + 1$

A-10. If $(\sqrt{3} + i)^{100} = 2^{99}\,(a + ib)$, then find

(i) $a^2 + b^2$ (ii) b

A-11. If n is a positive integer, prove the following

(i) $(1 + \cos\theta + i\sin\theta)^n + (1 + \cos\theta - i\sin\theta)^n = 2^{n+1}\cos^n\frac{\theta}{2}\cos\frac{n\theta}{2}$.

(ii) $(1 + i)^n + (1 - i)^n = 2^{\frac{n}{2}+1} . \cos\frac{n\pi}{4}$

A-12. Show that $e^{i2m\theta}\left(\frac{i\cot\theta+1}{i\cot\theta-1}\right)^m = 1$ $(m \in Z)$.

A-13. If $x_r = \cos\left(\frac{\pi}{3^r}\right) + i\sin\left(\frac{\pi}{3^r}\right)$, prove that $x_1 x_2 x_3$....... upto infinity = i.

Section (B) : Argument / Modulus / Conjugate Properties and Triangle Inequality

B-1. If z = x + iy is a complex number such that $z = (a + ib)^2$ then
(i) find $\bar{z}$, $\bar{z}$
(ii) show that $x^2 + y^2 = (a^2 + b^2)^2$ $x^2 + y^2 = (a^2 + b^2)^2$

B-2. If z_1 and z_2 are conjugate to each other, then find arg $(-z_1z_2)$.

B-3. If z ($\neq -1$) is a complex number such that $\frac{z-1}{z+1}$ is purely imaginary, then find |z|

B-4. If $|z - 2| = 2|z - 1|$, where z is a complex number, prove $|z|^2 = \frac{4}{3}$ Re (z) using

(i) polar form of z, (ii) z = x + iy, (iii) modulus, conjugate properties

B-5. For any two complex numbers z_1, z_2 and any two real numbers a, b show that
$|az_1 - bz_2|^2 + |bz_1 + az_2|^2 = (a^2 + b^2)(|z_1|^2 + |z_2|^2)$

B-6. If z_1 and z_2 are two complex numbers such that $|z_1| < 1 < |z_2|$ then prove that $\left|\frac{1-z_1\bar{z}_2}{z_1-z_2}\right| < 1$.

B-7. If k > 0, |z| = |w| = k and $\alpha = \frac{z-\bar{w}}{k^2+z\bar{w}}$, then find Re(α).

B-8. (i) If $w = \frac{z+i}{z-i}$ is purely real then find arg z.

(ii) If $w = \frac{z+4i}{z+2i}$ is purely imaginary then find |z + 3i|.

B-9. If $a = e^{i\alpha}$, $b = e^{i\beta}$, $c = e^{i\gamma}$ and $\cos\alpha + \cos\beta + \cos\gamma = 0 = \sin\alpha + \sin\beta + \sin\gamma$, then prove the following
(i) a + b + c = 0 (ii) ab + bc + ca = 0
(iii) $a^2 + b^2 + c^2 = 0$ (iv) $\Sigma\cos 2\alpha = 0 = \Sigma\sin 2\alpha$

B-10. If |z – 1 + i| + |z + i| = 1 then find range of principle argument of z.

Section (C) : Geometry of Complex Number and Rotation Theorem

C-1. If $|z - 2 + i| = 2$, then find the greatest and least value of $|z|$.

C-2. If $|z + 3| \leq 3$ then find minimum and maximum values of
(i) |z| (ii) |z – 1| (iii) |z + 1|

C-3. Interpret the following locus in $z \in C$.
(i) $1 < |z - 2i| < 3$ (ii) Im (z) ≥ 1

(iii) Arg (z – 3 – 4i) = $\pi/3$ (iv) $\text{Re}\left(\frac{z+2i}{iz+2}\right) \leq 4$ $(z \neq 2i)$

C-4. If O is origin and affixes of P, Q, R are respectively z, iz, z + iz. Locate the points on complex plane. If $\Delta PQR = 200$ then find
(i) $|z|$ (ii) sides of quadrilateral OPRQ

C-5. The three vertices of a triangle are represented by the complex numbers, 0, z_1 and z_2. If the triangle is equilateral, then show that $z_1^2 + z_2^2 = z_1z_2$. Further if z_0 is circumcentre then prove that $z_1^2 + z_2^2 = 3z_0^2$.

C-6. Let z_1 and z_2 be two roots of the equation $z^2 + az + b = 0$, z being complex. Further, assume that the origin, z_1 and z_2 form an equilateral triangle. Then show that $a^2 = 3b$.

C-7. Let $z_1 = 10 + 6i$ and $z_2 = 4 + 6i$. If z is any complex number such that the argument of $(z - z_1) / (z - z_2)$ is $\pi/4$, then find the length of arc of the locus.

C-8. Let I : $\text{Arg}\left(\dfrac{z-8i}{z+6}\right) = \pm\dfrac{\pi}{2}$

II : $\text{Re}\left(\dfrac{z-8i}{z+6}\right) = 0$

Show that locus of z in I or II lies on $x^2 + y^2 + 6x - 8y = 0$. Hence show that locus of z can also be represented by $\dfrac{z-8i}{z+6} + \dfrac{\bar{z}+8i}{\bar{z}+6} = 0$. Further if locus of z is expressed as $|z + 3 - 4i| = R$, then find R.

C-9. Show that $z\bar{z} + (4 - 3i)z + (4 + 3i)\bar{z} + 5 = 0$ represents circle. Hence find centre and radius.

C-10. If z_1 & z_2 are two complex numbers & if $\arg\dfrac{z_1 + z_2}{z_1 - z_2} = \dfrac{\pi}{2}$ but $|z_1 + z_2| \neq |z_1 - z_2|$ then identify the figure formed by the points represented by 0, z_1, z_2 & $z_1 + z_2$.

Section (D) : Cube root and n^th Root of Unity.

D-1. If ω ($\neq 1$) be a cube root of unity and $(1 + \omega^4)^n = (1 + \omega^2)^n$ then find the least positive integral value of n

D-2. When the polynomial $5x^3 + Mx + N$ is divided by $x^2 + x + 1$, the remainder is 0. Then find M + N.

D-3. Show that $(1 - \omega + \omega^2)(1 - \omega^2 + \omega^4)(1 - \omega^4 + \omega^8)$ .... to 2n factors $= 2^{2n}$

D-4. Let ω is non-real root of $x^3 = 1$

(i) If $P = \omega^n$, $(n \in N)$ and $Q = ({}^{2n}C_0 + {}^{2n}C_3 + ...) + ({}^{2n}C_1 + {}^{2n}C_4 + ...)\omega + ({}^{2n}C_2 + {}^{2n}C_5 + ...)\omega^2$ then find $\dfrac{P}{Q}$.

(ii) If $P = 1 - \dfrac{\omega}{2} + \dfrac{\omega^2}{4} - \dfrac{\omega^3}{8}$...... upto ∞ terms and $Q = \dfrac{1-\omega^2}{2}$ then find value of PQ.

D-5. If $x = 1 + i\sqrt{3}$; $y = 1 - i\sqrt{3}$ and z = 2, then prove that $x^p + y^p = z^p$ for every prime p > 3.

D-6. Solve $(z - 1)^4 - 16 = 0$. Find sum of roots. Locate roots, sum of roots and centroid of polygon formed by roots in complex plane.

D-7. Find the value(s) of the following

(i) $\left(\dfrac{1}{2} + \dfrac{\sqrt{-3}}{2}\right)^3$ (ii) $\left(\dfrac{1}{2} + \dfrac{\sqrt{-3}}{2}\right)^{3/4}$

Hence find continued product if two or more distinct values exists.

D-8. If 1, α_1, α_2, α_3, α_4 be the roots of $x^5 - 1 = 0$, then find the value of $\dfrac{\omega - \alpha_1}{\omega^2 - \alpha_1} \cdot \dfrac{\omega - \alpha_2}{\omega^2 - \alpha_2} \cdot \dfrac{\omega - \alpha_3}{\omega^2 - \alpha_3} \cdot \dfrac{\omega - \alpha_4}{\omega^2 - \alpha_4}$ (where ω is imaginary cube root of unity.)

D-9. $a = \cos\frac{2\pi}{7} + i\sin\frac{2\pi}{7}$ then find the quadratic equation whose roots are $\alpha = a + a^2 + a^4$ and $\beta = a^3 + a^5 + a^6$

PART - II : ONLY ONE OPTION CORRECT TYPE

Section (A) : Algebra of Complex Numbers and Its Representation and Demoivre's Theorem

A-1. If z is a complex number such that $|z| = 4$ and $\arg(z) = \frac{5\pi}{6}$, then z is equal to

(A) $-2\sqrt{3} + 2i$ (B) $2\sqrt{3} + i$ (C) $2\sqrt{3} - 2i$ (D) $-\sqrt{3} + i$

A-2. The complex numbers sin x + i cos 2x and cos x – i sin 2x are conjugate to each other, for

(A) $x = n\pi$ (B) $x = 0$ (C) $x = \frac{n\pi}{2}$ (D) no value of x

A.3. The least value of n ($n \in N$), for which $\left(\frac{1+i}{1-i}\right)^n$ is real, is

(A) 1 (B) 2 (C) 3 (D) 4

A-4. In G.P. the first term & common ratio are both $\frac{1}{2}\left(\sqrt{3}+i\right)$, then the modulus of n^{th} term is :

(A) 1 (B) 2^n (C) 4^n (D) 3^n

A-5. If $z = (3 + 7i)(p + iq)$, where $p, q \in I - \{0\}$, is purely imaginary, then minimum value of $|z|^2$ is

(A) 0 (B) 58 (C) $\frac{3364}{3}$ (D) 3364

A-6. If $z = x + iy$ and $z^{1/3} = a - ib$ then $\frac{x}{a} - \frac{y}{b} = k\left(a^2 - b^2\right)$ where k =

(A) 1 (B) 2 (C) 3 (D) 4

A-7. If $z = \frac{\pi}{4}(1+i)^4\left(\frac{1-\sqrt{\pi}\,i}{\sqrt{\pi}+i} + \frac{\sqrt{\pi}-i}{1+\sqrt{\pi}\,i}\right)$, then $\left(\frac{|z|}{\text{amp}(z)}\right)$ equals

(A) 1 (B) π (C) 3π (D) 4

A-8. The set of values of $a \in R$ for which $x^2 + i(a-1)x + 5 = 0$ will have a pair of conjugate imaginary roots is

(A) R (B) {1}
(C) $\{a : a^2 - 2a + 21 > 0\}$ (D) {0}

A-9. Let z is a complex number satisfying the equation, $z^3 - (3 + i)z + m + 2i = 0$, where $m \in R$. Suppose the equation has a real root α, then find the value of $\alpha^4 + m^4$

(A) 32 (B) 16 (C) 8 (D) 64

A-10. The expression $\left(\frac{1+i\tan\alpha}{1-i\tan\alpha}\right)^n - \frac{1+i\tan n\alpha}{1-i\tan n\alpha}$ when simplified reduces to :

(A) zero (B) $2\sin n\alpha$ (C) $2\cos n\alpha$ (D) none

A-11. If $(\cos\theta + i\sin\theta)(\cos 2\theta + i\sin 2\theta) \ldots (\cos n\theta + i\sin n\theta) = 1$, then the value of θ is

(A) $\frac{3m\pi}{n(n+1)}$, $m \in Z$ (B) $\frac{2m\pi}{n(n+1)}$, $m \in Z$ (C) $\frac{4m\pi}{n(n+1)}$, $m \in Z$ (D) $\frac{m\pi}{n(n+1)}$, $m \in Z$

A-12. Let principle argument of complex number be re-defined between $(\pi, 3\pi]$, then sum of principle arguments of roots of equation $z^6 + z^3 + 1 = 0$ is

(A) 0 (B) 3π (C) 6π (D) 12π

Section (B) : Argument / Modulus / Conjugate Properties and Triangle Inequality

B-1. If |z| = 1 and $\omega = \frac{z-1}{z+1}$ (where $z \neq -1$), the Re(ω) is

(A) 0 (B) $-\frac{1}{|z+1|^2}$ (C) $\left|\frac{z}{z+1}\right| \cdot \frac{1}{|z+1|^2}$ (D) $\frac{\sqrt{2}}{|z+1|^2}$

B-2. If $a^2 + b^2 = 1$, then $\frac{(1 + b + ia)}{(1 + b - ia)} =$

(A) 1 (B) 2 (C) b + ia (D) a + ib

B-3. If (2 + i)(2 + 2i) (2 + 3i) ...... (2 + ni) = x + iy, then the value of 5.8.13. ....($4 + n^2$)

(A) $(x^2 + y^2)$ (B) $\sqrt{(x^2+y^2)}$ (C) $2(x^2 + y^2)$ (D) (x + y)

B-4. If z = x + iy satisfies amp (z – 1) = amp (z + 3) then the value of (x – 1) : y is equal to

(A) 2 : 1 (B) 1 : 3 (C) – 1 : 3 (D) does not exist

B-5. If $z_1 = -3 + 5i$; $z_2 = -5 - 3i$ and z is a complex number lying on the line segment joining z_1 & z_2, then arg(z) can be :

(A) $-\frac{3\pi}{4}$ (B) $-\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{5\pi}{6}$

B-6. If $(1 + i)z = (1 - i)\bar{z}$ then z is

(A) t(1 – i), $t \in R$ (B) t (1 + i), $t \in R$ (C) $\frac{t}{1+i}$, $t \in R^+$ (D) $\frac{t}{1-i}$, $t \in R^+$

B-7. Let z and ω be two non-zero complex numbers such that $|z| = |\omega|$ and arg z = π – arg ω, then z equals

(A) ω (B) $-\omega$ (C) $\bar{\omega}$ (D) $-\bar{\omega}$

B-8. If z_1 and z_2 are two non-zero complex numbers such that $\left|\frac{z_1}{z_2}\right| = 2$ and $\arg(z_1z_2) = \frac{3\pi}{2}$, then $\frac{\bar{z}_1}{z_2}$ is equal to

(A) 2 (B) –2 (C) –2i (D) 2i

B-9. Number of complex numbers z such that |z| = 1 and $|z/\bar{z} + \bar{z}/z| = 1$ is (arg(z) $\in [0, 2\pi]$)

(A) 4 (B) 6 (C) 8 (D) more than 8

B-10. If $|z_1| = |z_2|$ and arg $(z_1/z_2) = \pi$, then $z_1 + z_2$ is equal to

(A) 1 (B) 3 (C) 0 (D) 2

B-11. The number of solutions of the system of equations Re $(z^2) = 0$, |z| = 2 is

(A) 4 (B) 3 (C) 2 (D) 1

B-12. If $|z^2 - 1| = |z|^2 + 1$, then z lies on :

(A) the real axis (B) the imaginary axis (C) a circle (D) an ellipse

B-13. If $|z - 2i| + |z - 2| \geq ||z| - |z - 2 - 2i||$, then locus of z is

(A) circle (B) line segment (C) point (D) complete x-y plane

Section (C) : Geometry of Complex Number and Rotation Theorem

C-1. The complex number z = x + iy which satisfy the equation $\left|\frac{z-5i}{z+5i}\right| = 1$ lie on :

(A) the x-axis (B) the straight line y = 5
(C) a circle passing through the origin (D) the y-axis

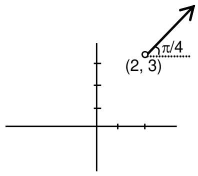
C-2. The inequality |z – 4| < | z – 2| represents :

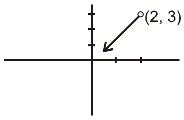
(A) Re(z) > 0 (B) Re(z) < 0 (C) Re (z) > 2 (D) Re(z) > 3

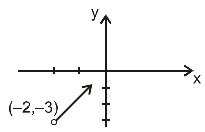
C-3. Let A, B, C represent the complex numbers z_1, z_2, z_3 respectively on the complex plane. If the circumcentre of the triangle ABC lies at the origin, then the orthocentre is represented by the complex number :

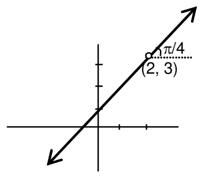
(A) $z_1 + z_2 - z_3$ (B) $z_2 + z_3 - z_1$ (C) $z_3 + z_1 - z_2$ (D) $z_1 + z_2 + z_3$

C-4. If Arg $(z - 2 - 3i) = \frac{\pi}{4}$, then the locus of z is

(A) 

(B) 

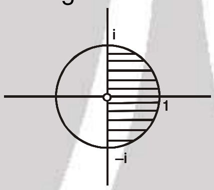
(C) 

(D) 

C-5. The system of equations $\left.\begin{array}{ll} |z+1-i| & = 2 \\ \text{Re } z & \geq 1 \end{array}\right\}$, where z is a complex number has :

(A) no solution (B) exactly one solution
(C) two distinct solutions (D) infinite solution

C-6. The locus of z which lies in shaded region is best represented by



(A) $|z| \leq 1, \ \frac{-\pi}{2} \leq \arg z \leq \frac{\pi}{2}$ (B) $|z| = 1, \ \frac{-\pi}{2} \leq \arg z \leq 0$

(C) $|z| \geq 0, \ 0 \leq \arg z \leq \frac{\pi}{2}$ (D) $|z| \leq 1, \ \frac{-\pi}{2} \leq \arg z \leq \pi$

C-7. The equation $|z - 1|^2 + |z + 1|^2 = 2$ represents
(A) a circle of radius '1' (B) a straight line
(C) the ordered pair (0, 0) (D) None of these

C-8. If $\arg\left(\frac{z-2}{z-4}\right) = \frac{\pi}{3}$ then locus of z is :

(A) equileteral triangle (B) arc of circle

(C) arc of ellipse (D) two rays making angle $\frac{\pi}{3}$ between them

C-9. The region of Argand diagram defined by $|z - 1| + |z + 1| \leq 4$ is :
(A) interior of an ellipse (B) exterior of a circle
(C) interior and boundary of an ellipse (D) exterior of ellipse

C-10. The vector $z = -4 + 5i$ is turned counter clockwise through an angle of 180º & stretched 1.5 times. The complex number corresponding to the newly obtained vector is :

(A) $6 - \frac{15}{2}i$ (B) $-6 + \frac{15}{2}i$ (C) $6 + \frac{15}{2}i$ (D) $-6 - \frac{15}{2}i$

C-11. The points z_1, z_2, z_3, z_4 in the complex plane are the vertices of a parallelogram taken in order if and only if :

(A) $z_1 + z_4 = z_2 + z_3$ (B) $z_1 + z_3 = z_2 + z_4$ (C) $z_1 + z_2 = z_3 + z_4$ (D) $z_1 z_3 = z_2 z_4$

C-12. Complex numbers z_1, z_2, z_3 are the vertices A, B, C respectively of an isosceles right angled triangle with right angle at C and $(z_1 - z_2)^2 = k(z_1 - z_3)(z_3 - z_2)$, then find k

(A) 1 (B) 2 (C) 3 (D) –2

C-13. If z_1, z_2, z_3 are vertices of an equilateral triangle inscribed in the circle $|z| = 2$ and if $z_1 = 1 + i\sqrt{3}$, then
(A) $z_2 = -2, z_3 = 1 + i\sqrt{3}$ (B) $z_2 = 2, z_3 = 1 - i\sqrt{3}$
(C) $z_2 = -2, z_3 = 1 - i\sqrt{3}$ (D) $z_2 = 1 - i\sqrt{3}, z_3 = -1 - i\sqrt{3}$

Section (D) : Cube root of unity and n^th Root of Unity.

D-1. Let z_1 and z_2 be two non real complex cube roots of unity and $|z - z_1|^2 + |z - z_2|^2 = \lambda$ be the equation of a circle with z_1, z_2 as ends of a diameter then the value of λ is
(A) 4 (B) 3 (C) 2 (D) $\sqrt{2}$

D-2. If $x = a + b + c$, $y = a\alpha + b\beta + c$ and $z = a\beta + b\alpha + c$, where α and β are imaginary cube roots of unity, then xyz =
(A) $2(a^3 + b^3 + c^3)$ (B) $2(a^3 - b^3 - c^3)$ (C) $a^3 + b^3 + c^3 - 3abc$ (D) $a^3 - b^3 - c^3$

D-3. If $1, \omega, \omega^2$ are the cube roots of unity, then $\Delta = \begin{vmatrix} 1 & \omega^n & \omega^{2n} \\ \omega^n & \omega^{2n} & 1 \\ \omega^{2n} & 1 & \omega^n \end{vmatrix}$, $(n \in I)$ is equal to-
(A) 0 (B) 1 (C) ω (D) ω^2

D-4. If $x^2 + x + 1 = 0$, then the numerical value of
$$\left(x+\frac{1}{x}\right)^2 + \left(x^2+\frac{1}{x^2}\right)^2 + \left(x^3+\frac{1}{x^3}\right)^2 + \left(x^4+\frac{1}{x^4}\right)^2 + \ldots\ldots + \left(x^{27}+\frac{1}{x^{27}}\right)^2$$ is equal to
(A) 54 (B) 36 (C) 27 (D) 18

D-5. If $a = 1 + \frac{x^3}{3!} + \frac{x^6}{6!} + \ldots\ldots$, $b = x + \frac{x^4}{4!} + \frac{x^7}{7!} + \ldots\ldots$, $c = \frac{x^2}{2!} + \frac{x^5}{5!} + \frac{x^8}{8!} + \ldots\ldots$
then find $a^3 + b^3 + c^3 - 3abc$.
(A) 1 (B) 2 (C) 3 (D) 4

D-6. If equation $(z - 1)^n = z^n = 1 (n \in N)$ has solutions, then n can be :
(A) 2 (B) 3 (C) 6 (D) 9

D-7. If α is non real and $\alpha = \sqrt[5]{1}$ then the value of $2^{|1 + \alpha + \alpha^2 + \alpha^{-2} - \alpha^{-1}|}$ is equal to
(A) 4 (B) 2 (C) 1 (D) 8

D-8. If $\alpha = e^{i8\pi/11}$ then Real $(\alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5)$ equals to :
(A) (B) 1 (C) $-\frac{1}{2}$ (D) -1

PART - III : MATCH THE COLUMN

1. **Match the column**

Column – I (Complex number Z)	**Column – II (Principal argument of Z)**
(A) $Z = \dfrac{(1+i)^5\left(1+\sqrt{3}i\right)^2}{-2i\left(-\sqrt{3}+i\right)}$.	(p) π
(B) $Z = \sin\dfrac{6\pi}{5} + i\left(1+\cos\dfrac{6\pi}{5}\right)$ is	(q) $-\dfrac{7\pi}{18}$
(C) $Z = 1 + \cos\left(\dfrac{11\pi}{9}\right) + i\sin\left(\dfrac{11\pi}{9}\right)$	(r) $\dfrac{9\pi}{10}$
(D) $Z = \sin x \sin(x - 60º) \sin(x + 60º)$ where $x \in \left(0, \dfrac{\pi}{3}\right)$ and $x \in R$	(s) $-\dfrac{5\pi}{12}$
	(t) 0

2.✎ **Column I** | **Column II (one of the values of z)**

	Column I		Column II
(A)	$z^4 - 1 = 0$	p.	$z = \cos\frac{\pi}{8} + i\sin\frac{\pi}{8}$
(B)	$z^4 + 1 = 0$	q.	$z = \cos\frac{\pi}{8} - i\sin\frac{\pi}{8}$
(C)	$iz^4 + 1 = 0$	r.	$z = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4}$
(D)	$iz^4 - 1 = 0$	s.	$z = \cos 0 + i\sin 0$

3. Which of the condition/ conditions in column II are satisfied by the quadrilateral formed by z_1, z_2, z_3, z_4 in order given in column I ?

	Column - I		**Column-II**
(A)	Parallelogram	(p)	$z_1 - z_4 = z_2 - z_3$
(B)	Rectangle	(q)	$\lvert z_1 - z_3\rvert = \lvert z_2 - z_4\rvert$
(C)	Rhombus	(r)	$\frac{z_1 - z_2}{z_3 - z_4}$ is real
(D)	Square	(s)	$\frac{z_1 - z_3}{z_2 - z_4}$ is purely imaginary
		(t)	$\frac{z_1 - z_2}{z_3 - z_2}$ is purely imaginary

4.✎ Let z_1 lies on $|z| = 1$ and z_2 lies on $|z| = 2$.

	Column – I		**Column – II**
(A)	Maximum value of $\lvert z_1 + z_2\rvert$	(p)	3
(B)	Minimum value of $\lvert z_1 - z_2\rvert$	(q)	1
(C)	Minimum value of $\lvert 2z_1 + 3z_2\rvert$	(r)	4
(D)	Maximum value of $\lvert z_1 - 2z_2\rvert$	(s)	5

Exercise-2

✎ Marked questions are recommended for Revision.

PART - I : ONLY ONE OPTION CORRECT TYPE

1. $\sin^{-1}\left\{\frac{1}{i}(z-1)\right\}$, where z is nonreal, can be the angle of a triangle if

(A) Re(z) = 1, Im(z) = 2 (B) Re(z) = 1, 0 < Im (z) ≤ 1
(C) Re(z) + Im(z) = 0 (D) Re(z) = 2, 0 < Im (z) ≤ 1

2. If $|z|^2 - 2iz + 2c(1 + i) = 0$, then the value of z is, where c is real.

(A) $z = c + 1\ i(-1 \pm \sqrt{1-2c-c^2})$, where $c \in [-1 - \sqrt{2}, -1 + \sqrt{2}]$
(B) $z = c - 1\ i(-1 \pm \sqrt{1-2c-c^2})$, where $c \in [-1 - \sqrt{2}, -1 + \sqrt{2}]$
(C) $z = 2c + 1\ i(-1 \pm \sqrt{1-2c-c^2})$, where $c \in [-1 - \sqrt{2}, -1 + \sqrt{2}]$
(D) $z = c + 1\ i(-1 \pm)\sqrt{1-2c-c^2}$, where $c \in [-1 - \sqrt{2}, 1 + \sqrt{2}]$

3.✎ If $(a + ib)^5 = \alpha + i\beta$, then $(b + ia)^5$ is equal to

(A) $\beta + i\alpha$ (B) $\alpha - i\beta$ (C) $\beta - i\alpha$ (D) $-\alpha - i\beta$

4. Let z be non real number such that $\dfrac{1+z+z^2}{1-z+z^2} \in R$, then value of $7|z|$ is
(A) 1 (B) 3 (C) 5 (D) 7

5. If $|z_1| = 2$, $|z_2| = 3$, $|z_3| = 4$ and $|z_1 + z_2 + z_3| = 2$, then the value of $|4z_2z_3 + 9z_3z_1 + 16z_1z_2|$
(A) 24 (B) 48 (C) 96 (D) 120

6. The minimum value of $|3z-3| + |2z-4|$ equal to
(A) 1 (B) 2 (C) 3 (D) 4

7. If $|z_1 - 1| < 1$, $|z_2 - 2| < 2$, $|z_3 - 3| < 3$, then $|z_1 + z_2 + z_3|$
(A) is less than 6
(B) is more than 3
(C) is less than 12
(D) lies between 6 and 12

8. Let O = (0, 0) ; A=(3, 0) ; B = (0, –1) and C=(3, 2),then minimum value of $|z|+|z-3|+|z+i|+|z-3-2i|$ occur at
(A) intersection point of AB and CO
(B) intersection point of AC and BO
(C) intersection point of CB and AO
(D) mean of O, A, B, C

9. Given z is a complex number with modulus 1. Then the equation $[(1 + ia)/(1 - ia)]^4 = z$ in 'a' has
(A) all roots real and distinct
(B) two real and two imaginary
(C) three roots real and one imaginary
(D) one root real and three imaginary

10. The real values of the parameter 'a' for which at least one complex number z = x + iy satisfies both the equality $|z - ai| = a + 4$ and the inequality $|z - 2| < 1$.
(A) $\left(-\dfrac{21}{10}, -\dfrac{5}{6}\right)$ (B) $\left(-\dfrac{7}{2}, -\dfrac{5}{6}\right)$ (C) $\left(\dfrac{5}{6}, \dfrac{7}{2}\right)$ (D) $\left(-\dfrac{21}{10}, \dfrac{7}{2}\right)$

11. The points of intersection of the two curves $|z - 3| = 2$ and $|z| = 2$ in an argand plane are:
(A) $\dfrac{1}{2}\left(7 \pm i\sqrt{3}\right)$ (B) $\dfrac{1}{2}\left(3 \pm i\sqrt{7}\right)$ (C) $\dfrac{3}{2} \pm i\sqrt{\dfrac{7}{2}}$ (D) $\dfrac{7}{2} \pm i\sqrt{\dfrac{3}{2}}$

12. The equation of the radical axis of the two circles represented by the equations, $|z - 2| = 3$ and $|z - 2 - 3i| = 4$ on the complex plane is :
(A) $3iz - 3i\bar{z} - 2 = 0$ (B) $3iz - 3i\bar{z} + 2 = 0$ (C) $iz - i\bar{z} + 1 = 0$ (D) $2iz - 2i\bar{z} + 3 = 0$

13. If $\log_{1/2}\left(\dfrac{|z-1| + 4}{3|z-1| - 2}\right) > 1$, then the locus of z is
(A) Exterior to circle with center 1 + i0 and radius 10
(B) Interior to circle with center 1 + i0 and radius 10
(C) Circle with center 1 + i0 and radius 10
(D) Circle with center 2 + i0 and radius 10

14. Points z_1 & z_2 are adjacent vertices of a regular octagon. The vertex z_3 adjacent to z_2 ($z_3 \neq z_1$) is represented by :
(A) $z_2 + \dfrac{1}{\sqrt{2}}(1 \pm i)(z_1 + z_2)$
(B) $z_2 + \dfrac{1}{\sqrt{2}}(1 \pm i)(z_1 - z_2)$
(C) $z_2 + \dfrac{1}{\sqrt{2}}(1 \pm i)(z_2 - z_1)$
(D) none of these

15. If $p = a + b\omega + c\omega^2$; $q = b + c\omega + a\omega^2$ and $r = c + a\omega + b\omega^2$ where a, b, c ≠ 0 and ω is the non-real complex cube root of unity, then :
(A) $p + q + r = a + b + c$
(B) $p^2 + q^2 + r^2 = a^2 + b^2 + c^2$
(C) $p^2 + q^2 + r^2 = 2(pq + qr + rp)$
(D) None of these

16. The points $z_1 = 3 + \sqrt{3}\,i$ and $z_2 = 2\sqrt{3} + 6\,i$ are given on a complex plane. The complex number lying on the bisector of the angle formed by the vectors z_1 and z_2 is :

(A) $z = \frac{(3+2\sqrt{3})}{2} + \frac{\sqrt{3}+2}{2}i$ (B) $z = 5 + 5\,i$

(C) $z = -1 - i$ (D) none

17. Let ω be the non real cube root of unity which satisfy the equation $h(x) = 0$ where $h(x) = x\,f(x^3) + x^2g(x^3)$. If $h(x)$ is polynomial with real coefficient then which statement is incorrect.

(A) $f(1) = 0$ (B) $g(1) = 0$ (C) $h(1) = 0$ (D) $g(1) \neq f(1)$

18. If $1, \alpha_1, \alpha_2, \alpha_3, \ldots\ldots, \alpha_{n-1}$ be the n^{th} roots of unity, then the value of

$\sin\frac{\pi}{n} . \sin\frac{2\pi}{n} . \sin\frac{3\pi}{n} \ldots\ldots \sin\frac{(n-1)\pi}{n}$ equals

(A) $\frac{n}{2^n}$ (B) $\frac{n}{2^{n-1}}$ (C) $\frac{n+1}{2^{n-1}}$ (D) $\frac{n}{2^{n+1}}$

PART - II : NUMERICAL VALUE QUESTIONS

INSTRUCTION :

- The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto two digit.
- If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal placed.

1. If a and b are positive integer such that $N = (a + ib)^3 - 107i$ is a positive integer then find the value of $\frac{N}{5}$

2. Let z,w be complex numbers such that $\bar{z} + i\bar{w} = 0$ and arg zw = π. If Re(z) < 0 and principal arg(z) is

3. If $x = 9^{1/3}\, 9^{1/9}\, 9^{1/27} \ldots\infty$, $y = 4^{1/3}\, 4^{-1/9}\, 4^{1/27} \ldots \infty$, and $z = \sum_{r=1}^{\infty}(1+i)^{-r}$ and principal argument of $P = (x + yz)$ is $-\tan^{-1}\left(\frac{\sqrt{a}}{b}\right)$ then determine $a^2 + b^2$. (where a & b are co-prime natural numbers)

4. $z_1, z_2 \in c$ and $z_1^2 + z_2^2 \in R$, $z_1(z_1^2 - 3z_2^2) = 2$, $z_2(3z_1^2 - z_2^2) = 11$. If $z_1^2 + z_2^2 = \lambda$ then determine λ

5. Let $|z| = 2$ and $w = \frac{z+1}{z-1}$ where z, w$\in$C (where C is the set of complex numbers). If maximum and minimum value of |w| is M and m respectively then value of M + m.

6. A function 'f' is defined by $f(z) = (4 + i)z^2 + \alpha z + \gamma$ for all complex number z, where α and γ are complex numbers if f(1) and f(i) are both real and the smallest possible values of $|\alpha| + |\gamma|$ is p then determine p.

7. If z and ω are two non-zero complex numbers such that $|z\omega| = 1$, and $\arg(z) - \arg(\omega) = \frac{\pi}{2}$, then find the value of $10\, i\, \bar{z}\, \omega$.

8. Number of complex number satisfying $|z| = \max\{|z - 1|, |z + 1|\}$.

9. If z_1 & z_2 both satisfy the relation, $z + \bar{z} = 2\,|z - 1|$ and $\arg(z_1 - z_2) = \frac{\pi}{4}$, then find the imaginary part of $(z_1 + z_2)$.

10. If $a_1, a_2, a_3 \ldots a_n, A_1, A_2, A_3 \ldots A_n$, k are all real numbers and number of imaginary roots of the equation $\frac{A_1^2}{x - a_1} + \frac{A_2^2}{x - a_2} + \ldots\ldots + \frac{A_n^2}{x - a_n} = k$ is α (where all $A_i \neq 0$). Then find the value of $\alpha + 15$.

11. How many complex number z such that $|z| < \frac{1}{3}$ and $\sum_{r=1}^{n} a_r z^r = 1$ where $|a_r| < 2$.

12. If a variable circle S touches $S_1 : |z - z_1| = 7$ internally and $S_2 : |z - z_2| = 4$ externally while the curves S_1 & S_2 touch internally to each other, $(z_1 \neq z_2)$. If the eccentricity of the locus of the centre of the curve S is 'e' find the value of e.

13. Given that, $|z - 1| = 1$, where ' z ' is a point on the argand plane. $\frac{z-2}{2z} = \alpha\ i \tan(\arg z)$. Then determine $\frac{1}{\alpha^4}$.

14. Area of the region formed by $|z| \le 4$ & $-\frac{\pi}{2} \le \arg z \le \frac{\pi}{3}$ on the Argand diagram is

15. The points A, B, C represent the complex numbers z_1, z_2, z_3 respectively on a complex plane & the angle B & C of the triangle ABC are each equal to $\frac{1}{2}(\pi - \alpha)$. If $(z_2 - z_3)^2 = \lambda\,(z_3 - z_1)(z_1 - z_2)\sin^2\frac{\alpha}{2}$ then determine λ^2

16. If ω and ω^2 are the non-real cube roots of unity and a, b, c $\in$ R such that $\frac{1}{a+\omega} + \frac{1}{b+\omega} + \frac{1}{c+\omega} = 2\omega^2$ and $\frac{1}{a+\omega^2} + \frac{1}{b+\omega^2} + \frac{1}{c+\omega^2} = 2\omega$. If $\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} = \lambda$ then determine λ^4

17. If $L = \lim_{n\to\infty}\left[\frac{n}{(1-n\omega)(1-n\omega^2)} + \frac{n}{(2-n\omega)(2-n\omega^2)} + \ldots\ldots + \frac{n}{(n-n\omega)(n-n\omega^2)}\right]$ then value of $9L^2$ is

18. The value of $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7} - i\cos\frac{2\pi k}{7}\right) = \alpha$ then find α^4

19. If $Z_r = \left(e^{i\frac{2\pi}{15}}\right)^r$ then value of $\arg\left(\frac{1 + Z_1 + Z_2 + Z_3 + \ldots\ldots\ldots + Z_7}{1 + Z_8 + Z_9 + Z_{10} + \ldots\ldots. + Z_{14}}\right)$ is

20. If $A_1, A_2, \ldots\ldots., A_n$ be the vertices of an n-sided regular polygon such that $\frac{1}{A_1A_2} = \frac{1}{A_1A_3} + \frac{1}{A_1A_4}$, then find the value of n

PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

1. If the biquadratic $x^4 + ax^3 + bx^2 + cx + d = 0$ (a, b, c, d $\in$ R) has 4 non real roots, two with sum 3 + 4i and the other two with product 13 + i.
(A) b = 51 (B) a = –6 (C) c = –70 (D) d = 170

2. The quadratic equation $z^2 + (p + ip')z + q + iq' = 0$; where p, p′, q, q′ are all real.
(A) if the equation has one real root then $q'^2 - pp'q' + qp'^2 = 0$.
(B) if the equation has two equal roots then $pp' = 2q'$.
(C) if the equation has two equal roots then $p^2 - p'^2 = 4q$
(D) if the equation has one real root then $p'^2 - pp'q' + q'^2 = 0$.

3. The value of $i^n + i^{-n}$, for $i = \sqrt{-1}$ and $n \in I$ is :
(A) $\frac{2^n}{(1-i)^{2n}} + \frac{(1+i)^{2n}}{2^n}$ (B) $\frac{(1+i)^{2n}}{2^n} + \frac{(1-i)^{2n}}{2^n}$ (C) $\frac{(1+i)^{2n}}{2^n} + \frac{2^n}{(1-i)^{2n}}$ (D) $\frac{2^n}{(1+i)^{2n}} + \frac{2^n}{(1-i)^{2n}}$

4. If amp $(z_1z_2) = 0$ and $|z_1| = |z_2| = 1$, then
(A) $z_1 + z_2 = 0$ (B) $z_1z_2 = 1$ (C) $z_1 = \bar{z}_2$ (D) $z_1 = z_2$

5. Let z_1 and z_2 are two complex numbers such that $(1 - i)z_1 = 2z_2$ and $\arg(z_1z_2) = \frac{\pi}{2}$, then $\arg(z_2)$ is equal to

(A) $3\pi/8$ (B) $\pi/8$ (C) $5\pi/8$ (D) $-7\pi/8$

6. If $|z_1 + z_2|^2 = |z_1|^2 + |z_z|^2$ (where z_1 and z_2 are non-zero complex numbers), then

(A) $\frac{z_1}{z_2}$ is purely real (B) $\frac{z_1}{z_2}$ is purely imaginary

(C) $z_1\bar{z}_2 + z_2\bar{z}_1 = 0$ (D) amp $\frac{z_1}{z_2}$ may be equal to $\frac{\pi}{2}$

7. a, b, c are real numbers in the polynomial, $P(z) = 2z^4 + az^3 + bz^2 + cz + 3$. If two roots of the equation $P(z) = 0$ are 2 and i. Then which of the following are true.

(A) $a = -\frac{11}{2}$ (B) $b = 5$ (C) $c = -\frac{11}{2}$ (D) $a = -11$

8. If $Z = \frac{(1+i)(1+2i)(1+3i)......(1+ni)}{(1-i)(2-i)(3-i)......(n-i)}$, $n \in N$ then principal argument of Z can be

(A) 0 (B) $\frac{\pi}{2}$ (C) $-\frac{\pi}{2}$ (D) π

9. For complex numbers z and w, if $|z|^2w - |w|^2 z = z - w$. Which of the following can be true :

(A) $z = w$ (B) $z\bar{w} = 1$ (C) $|z| = |w| = 2, z \neq w$ (D) $\bar{z}w = 1$

10. If z satisfies the inequality $|z - 1 - 2i| \leq 1$, then which of the following are true.

(A) maximum value of $|z| = \sqrt{5}+1$ (B) minimum value of $|z| = \sqrt{5}-1$

(C) maximum value of $\arg(z) = \pi/2$ (D) minimum value of $\arg(z) = \tan^{-1}\left(\frac{3}{4}\right)$

11. The curve represented by $z = \frac{3}{2+\cos\theta+i\sin\theta}$, $\theta \in [0, 2\pi)$

(A) never meets the imaginary axis (B) meets the real axis in exactly two points

(C) has maximum value of |z| as 3 (D) has minimum value of |z| as 1

12. POQ is a straight line through the origin O . P and Q represent the complex number a + i b and c + i d respectively and OP = OQ. Then which of the follwoing are true :

(A) $|a + i b| = | c + i d|$ (B) $a + c = b + d$

(C) $\arg (a + i b) = \arg (c + i d)$ (D) none of these

13. Let $i = \sqrt{-1}$. Define a sequence of complex number by $z_1 = 0$, $z_{n+1} = z_n^2 + i$ for $n \geq 1$. Then which of the following are true.

(A) $|z_{2050}| = \sqrt{3}$ (B) $|z_{2017}| = \sqrt{2}$ (C) $|z_{2016}| = 1$ (D) $|z_{2111}| = \sqrt{2}$

14. If $|z_1| = |z_2| = ... = |z_n| = 1$ then which of the following are true.

(A) $\bar{z}_1 = \frac{1}{z_1}$ (B) $|z_1 + z_2 + ... + z_n| = \left|\frac{1}{z_1}+\frac{1}{z_2}+......+\frac{1}{z_n}\right|$.

(C) Centroid of polygon with 2n vertices $z_1, z_2, ... z_n$ $\frac{1}{z_1}, \frac{1}{z_2}, ... \frac{1}{z_n}$ (need not be in order) lies on real axis.

(D) Centroid of polygon with 2n vertices $z_1, z_2, ... z_n$ $\frac{1}{z_1}, \frac{1}{z_2}, ... \frac{1}{z_n}$ (need not be in order) lies on imaginary axis.

15. If $2\cos\theta = x + \frac{1}{x}$ and $2\cos\phi = y + \frac{1}{y}$, then which of the following statement can be true?

(A) $x^n + \frac{1}{x^n} = 2\cos(n\theta),\ n \in z$ (B) $\frac{x}{y} + \frac{y}{x} = 2\cos(\theta - \phi)$

(C) $xy + \frac{1}{xy} = 2\cos(\theta + \phi)$ (D) $x^m y^n + \frac{1}{x^m y^n} = 2\cos(m\theta + n\phi),\ m, n \in z$

16. If $\left|\frac{z-\alpha}{z-\beta}\right| = k,\ k > 0$ where, $z = x + iy$ and $\alpha = \alpha_1 + i\alpha_2$, $\beta = \beta_1 + i\beta_2$ are fixed complex numbers. Then which of the following are true

(A) if $k \neq 1$ then locus is a circle whose centre is $\left(\frac{k^2\beta - \alpha}{k^2 - 1}\right)$

(B) if $k \neq 1$ then locus is a circle whose radius is $\left|\frac{k(\alpha - \beta)}{1 - k^2}\right|$

(C) if $k = 1$ then locus is perpendicular bisector of line joining $\alpha = \alpha_1 + i\alpha_2$ and $\beta = \beta_1 + i\beta_2$

(D) if $k \neq 1$ then locus is a circle whose centre is $\left(\frac{k^2\alpha - \beta}{k^2 - 1}\right)$

17. The locus of equation $\text{Arg}\left(\frac{z-1-2i}{z+3+i}\right) = \frac{\pi}{3}$ represents part of circle in which

(A) centre is $\left[-1 - \frac{\sqrt{3}}{2} + i\left(\frac{1}{2} + \frac{2}{\sqrt{3}}\right)\right]$ (B) radius is $\frac{5}{\sqrt{3}}$

(C) centre is $\left[-1 - \frac{\sqrt{3}}{2} - i\left(\frac{1}{2} + \frac{2}{\sqrt{3}}\right)\right]$ (D) radius is $\frac{\sqrt{5}}{3}$

18. The equation $||z + i| - |z - i|| = k$ represents

(A) a hyperbola if $0 < k < 2$ (B) a pair of ray if $k > 2$

(C) a straight line if $k = 0$ (D) a pair of ray if $k = 2$

19. The equation $|z - i| + |z + i| = k,\ k > 0$, can represent

(A) an ellipse if $k > 2$ (B) line segment if $k = 2$ (C) an ellipse if $k = 5$ (D) no locus if $k = 1$

20. If $|z_1| = |z_2| = |z_3| = 1$ and z_1, z_2, z_3 are represented by the vertices of an equilateral triangle then

(A) $z_1 + z_2 + z_3 = 0$ (B) $z_1z_2z_3 = 1$ (C) $z_1z_2 + z_2z_3 + z_3z_1 = 0$ (D) $z_2^3 + z_3^3 = 2z_1^3$

21. Let z_1, z_2, z_3 be three distinct complex numbers satisfying, $|z_1-1| = |z_2-1| = |z_3-1| = 1$. Let A, B & C be the points representing vertices of equilateral triangle in the Argand plane corresponding to z_1, z_2 and z_3 respectively. Which of the following are true

(A) $z_1 + z_2 + z_3 = 3$ (B) $z_1^2 + z_2^2 + z_3^2 = 3$

(C) area of triangle ABC $= \frac{3\sqrt{3}}{4}$ (D) $z_1z_2 + z_2z_3 + z_3z_1 = 1$

22. If $1, \alpha_1, \alpha_2, \alpha_3, \ldots\ldots, \alpha_{n-1}$ be the n^{th} roots of unity, then which of the following are true

(A) $\frac{1}{1-\alpha_1} + \frac{1}{1-\alpha_2} + \ldots\ldots + \frac{1}{1-\alpha_{n-1}} = \frac{n-1}{2}$

(B) $(1 - \alpha_1)(1 - \alpha_2)(1 - \alpha_3)\ldots\ldots(1 - \alpha_{n-1}) = n.$

(C) $(2 - \alpha_1)(2 - \alpha_2)(2 - \alpha_3)\ldots\ldots(2 - \alpha_{n-1}) = 2^n - 1$

(D) $\frac{1}{1-\alpha_1} + \frac{1}{1-\alpha_2} + \ldots\ldots + \frac{1}{1-\alpha_{n-1}} = \frac{n}{2}$

23. Which of the following are true.

(A) $\cos x + {}^nC_1 \cos 2x + {}^nC_2 \cos 3x + ... + {}^nC_n \cos (n + 1) x = 2^n. \cos^n \frac{x}{2} . \cos \left(\frac{n+2}{2}\right) x$

(B) $\sin x + {}^nC_1 \sin 2x + {}^nC_2 \sin 3x + ... + {}^nC_n \sin (n + 1) x = 2^n. \cos^n \frac{x}{2} . \sin \left(\frac{n+2}{2}\right) x$

(C) $1 + {}^nC_1 \cos x + {}^nC_2 \cos 2x + ... + {}^nC_n \cos nx = 2^n. \cos^n \frac{x}{2} . \cos\left(\frac{nx}{2}\right)$

(D) ${}^nC_1 \sin x + {}^nC_2 \sin 2x + ... + {}^nC_n \sin n x = 2^n. \cos^n \frac{x}{2} . \sin\left(\frac{nx}{2}\right)$

24. If α, β, γ are distinct roots of $x^3 - 3x^2 + 3x + 7 = 0$ and ω is non-real cube root of unity, then the value of $\frac{\alpha-1}{\beta-1}+\frac{\beta-1}{\gamma-1}+\frac{\gamma-1}{\alpha-1}$ can be equal to :

(A) ω^2 (B) $2\omega^2$ (C) $3\omega^2$ (D) 3ω

25. If z is a complex number then the equation $z^2 + z\,|z| + |z^2| = 0$ is satisfied by
(ω and ω^2 are imaginary cube roots of unity)

(A) $z = k\,\omega$ where $k \in R$
(B) $z = k\,\omega^2$ where k is non negative real
(C) $z = k\,\omega$ where k is positive real
(D) $z = k\,\omega^2$ where $k \in R$.

26. If α is imaginary n^{th} $(n \geq 3)$ root of unity. Which of the following are true.

(A) $\sum_{r=1}^{n-1}(n-r)\,\alpha^r = \frac{n\alpha}{1-\alpha}$
(B) $\sum_{r=1}^{n-1}(n-r)\sin\frac{2r\pi}{n} = \frac{n}{2}\cot\frac{\pi}{n}$.
(C) $\sum_{r=1}^{n-1}(n-r)\,\cos\,\frac{2r\pi}{n} = -\frac{n}{2}$
(D) $\sum_{r=1}^{n-1}(n-r)\,\alpha^r = \frac{n}{1-\alpha}$

27. Which of the following is true?

(A) Number of roots of the equation $z^{10} - z^5 - 992 = 0$ with real part positive = 5
(B) Number of roots of the equation $z^{10} - z^5 - 992 = 0$ with real part negative = 5
(C) Number of roots of the equation $z^{10} - z^5 - 992 = 0$ with imaginary part non-negative = 6
(D) Number of roots of the equation $z^{10} - z^5 - 992 = 0$ with imaginary part negative = 4

PART - IV : COMPREHENSION

Comprehension # 1 (Q. No. 1 - 2)

Let $(1 + x)^n = C_0 + C_1x + C_2x^2 + ...+ C_nx^n$. For sum of series $C_0 + C_1 + C_2 + ...$, put x = 1. For sum of series $C_0 + C_2 + C_4 + C_6 + ...$, or $C_1 + C_3 + C_5 + ...$ add or substract equations obtained by putting x = 1 and x = – 1.

For sum of series $C_0 + C_3 + C_6 + ...$ or $C_1 + C_4 + C_7 + ...$ or $C_2 + C_5 + C_8 + ...$ we subsitute x = 1, $x = \omega$, $x = \omega^2$ and add or manipulate results.

Similarly, if suffixes differe by 'p' then we substitute p^{th} roots of unity and add.

1. $C_0 + C_3 + C_6 + C_9 +........ =$

(A) $\frac{1}{3}\left[2^n - 2\cos\frac{n\pi}{3}\right]$ (B) $\frac{1}{3}\left[2^n + 2\cos\frac{n\pi}{3}\right]$ (C) $\frac{1}{3}\left[2^n - 2\sin\frac{n\pi}{3}\right]$ (D) $\frac{1}{3}\left[2^n + 2\sin\frac{n\pi}{3}\right]$

2. $C_1 + C_5 + C_9 +.... =$

(A) $\frac{1}{4}\left[2^n - 2^{n/2}2\cos\frac{n\pi}{4}\right]$
(B) $\frac{1}{4}\left[2^n + 2^{n/2}2\cos\frac{n\pi}{4}\right]$
(C) $\frac{1}{4}\left[2^n - 2^{n/2}2\sin\frac{n\pi}{4}\right]$
(D) $\frac{1}{4}\left[2^n + 2^{n/2}2\sin\frac{n\pi}{4}\right]$

Comprehension # 2 (Q. No. 3 to 6)

As we know $e^{i\theta} = \cos\theta + i\sin\theta$ and $(\cos\theta_1 + i\sin\theta_1).(\cos\theta_2 + i\sin\theta_2) = \cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)$

Let $\alpha, \beta, \gamma \in R$ such that $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -\frac{3}{2}$

3. $\Sigma \sin(\alpha + \beta) = \Sigma \cos(\alpha + \beta) =$

(A) 0 (B) $3\cos\alpha \cos\beta \cos\gamma$ (C) $3\cos(\alpha + \beta + \gamma)$ (D) 3

4. $\Sigma\cos(2\alpha - \beta - \gamma)$

(A) 0 (B) $3\cos\alpha \cos\beta \cos\gamma$ (C) $3\cos(\alpha + \beta + \gamma)$ (D) 3

5. $\Sigma \cos 3\alpha =$

(A) 0 (B) $3\cos\alpha \cos\beta \cos\gamma$ (C) $3\cos(\alpha + \beta + \gamma)$ (D) 3

6. If $\theta \in R$ then $\dfrac{\Sigma\cos^3(\theta+\alpha)}{\Pi\cos(\theta+\alpha)} =$

(A) 0 (B) $3\cos\alpha \cos\beta \cos\gamma$ (C) $3\cos(\alpha + \beta + \gamma)$ (D) 3

Comprehension # 3(Q. No. 7 to 8)

ABCD is a rhombus. Its diagonals AC and BD intersect at the point M and satisfy BD = 2AC. Let the points D and M represent complex numbers 1 + i and 2 – i respecetively.

If θ is arbitary real, then $z = re^{i\theta}$, $R_1 \le r \le R_2$ lies in annular region formed by concentric circles $|z| = R_1$, $|z| = R_2$.

7. A possible representation of point A is

(A) $3 - \frac{i}{2}$ (B) $3 + \frac{i}{2}$ (C) $1 + \frac{3}{2}i$ (D) $3 - \frac{3}{2}i$

8. If z is any point on segment DM then $w = e^{iz}$ lies in annular region formed by concentric circles.

(A) $|w|_{min} = 1$, $|w|_{max} = 2$ (B) $|w|_{min} = \frac{1}{e}$, $|w|_{max} = e$

(C) $|w|_{min} = \frac{1}{e^2}$, $|w|_{max} = e^2$ (D) $|w|_{min} = \frac{1}{2}$, $|w|_{max} = 1$

Comprehension # 4 (Q. No. 9 to 10)

Logarithm of a complex number z = x + iy is given by

$\log_e z = \log_e(x + iy) = \log_e(|z|e^{i\theta})$ (where $\theta = \arg(z)$)

$= \log_e|z| + \log_e e^{i\theta}$

$= \log_e|z| + i\theta$

$= \log_e\sqrt{(x^2+y^2)} + i\arg(z)$

$\therefore$ $\log_e(z) = \log_e|z| + i\arg(z)$

In general $\log_e(x + iy) = \frac{1}{2}\log_e(x^2 + y^2) + i(2n\pi + \arg(x + iy))$ where $n \in I$.

9. Write $\log_e(1 + \sqrt{3}\, i)$ in (a + ib) form

(A) $\log_e 2 + i(2n\pi + \frac{\pi}{3})$ (B) $\log_e 3 + i(n\pi + \frac{\pi}{3})$ (C) $\log_e 2 + i(2n\pi + \frac{\pi}{6})$ (D) $\log_e 2 + i(2n\pi - \frac{\pi}{3})$

10. Find the real part of $(1 - i)^{-i}$.

(A) $e^{\pi/4 + 2n\pi}\cos\left(\frac{1}{2}\log_e 2\right)$ (B) $e^{-\pi/4 + 2n\pi}\cos\left(\frac{1}{2}\log_e 2\right)$

(C) $e^{-\pi/4 + 2n\pi}\cos(\log_e 2)$ (D) $e^{-\pi/2 + 2n\pi}\cos\left(\frac{1}{2}\log_e 2\right)$

Exercise-3

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

✍ Marked questions are recommended for Revision.
*** Marked Questions may have more than one correct option.**

1*.✍ Let z_1 and z_2 be two distinct complex numbers and let $z = (1 - t) z_1 + tz_2$ for some real number t with $0 < t < 1$. If Arg(w) denotes the principal argument of a nonzero complex number w, then **[IIT-JEE-2010, Paper-1, (3, 0)/84]**

(A) $|z - z_1| + |z - z_2| = |z_1 - z_2|$ (B) $\text{Arg}(z - z_1) = \text{Arg}(z - z_2)$

(C) $\begin{vmatrix} z - z_1 & \bar{z} - \bar{z}_1 \\ z_2 - z_1 & \bar{z}_2 - \bar{z}_1 \end{vmatrix} = 0$ (D) $\text{Arg}(z - z_1) = \text{Arg}(z_2 - z_1)$

2. Let ω be the complex number $\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$. Then the number of distinct complex numbers z satisfying $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0$ is equal to **[IIT-JEE-2010, Paper-1, (3, 0)/84]**

3.✍ Match the statements in **Column-I** with those in **Column-II.** **[IIT-JEE-2010, Paper-2, (8, 0)/79]**
[Note : Here z takes values in the complex plane and Im z and Re z denote, respectively, the imaginary part and the real part of z.]

Column-I	**Column-II**
(A) The set of points z satisfying $|z - i|z|| = |z + i|z||$ is contained in or equal to	(p) an ellipse with eccentricity $\frac{4}{5}$
(B) The set of points z satisfying $|z + 4| + |z - 4| = 10$ is contained in or equal to	(q) the set of points z satisfying Im z = 0
(C) If $|w| = 2$, then the set of points $z = w - \frac{1}{w}$ is contained in or equal to	(r) the set of point z satisfying $|\text{Im } z| \le 1$
(D) If $|w| = 1$, then the set of points $z = w + \frac{1}{w}$ is contained in or equal to	(s) the set of points z satisfying $|\text{Re } z| \le 2$
	(t) the set of points z satisfying $|z| \le 3$

4. If z is any complex number satisfying $|z - 3 - 2i| \le 2$, then the minimum value of $|2z - 6 + 5i|$ is **[IIT-JEE 2011, Paper-1, (4, 0), 80]**

5. Let $\omega = e^{\frac{\pi i}{3}}$, and a, b, c, x, y, z be non-zero complex numbers such that

$$a + b + c = x$$
$$a + b\omega + c\omega^2 = y$$
$$a + b\omega^2 + c\omega = z.$$

Then the value of $\frac{|x|^2 + |y|^2 + |z|^2}{|a|^2 + |b|^2 + |c|^2}$ is **[IIT-JEE 2011, Paper-2, (4, 0), 80**

6. Let z be a complex number such that the imaginary part of z is non zero and $a = z^2 + z + 1$ is real. Then a cannot take the value **[IIT-JEE 2012, Paper-1, (3, –1), 70]**

(A) –1 (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

7. Let complex numbers α and $\frac{1}{\bar{\alpha}}$ lies on circles $(x - x_0)^2 + (y - y_0)^2 = r^2$ and $(x - x_0)^2 + (y - y_0)^2 = 4r^2$, respectively. If $z_0 = x_0 + iy_0$ satisfies the equation $2|z_0|^2 = r^2 + 2$, then $|\alpha| =$

[JEE (Advanced) 2013, Paper-1, (2, 0)/60]

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{7}}$ (D) $\frac{1}{3}$

8.* Let $w = \frac{\sqrt{3}+i}{2}$ and $P = \{w^n : n = 1,2,3,....\}$. Further $H_1 = \left\{z \in C : \text{Re } z > \frac{1}{2}\right\}$ and $H_2 = \left\{z \in C : \text{Re } z < -\frac{1}{2}\right\}$, where C is the set of all complex numbers. If $z_1 \in P \cap H_1$, $z_2 \in P \cap H_2$ and O represents the origin, then $\angle z_1 O z_2 =$ **[JEE (Advanced) 2013, Paper-2, (3, –1)/60]**

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{6}$ (C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{6}$

9.* Let ω be a complex cube root of unity with $\omega \neq 1$ and $P = [p_{ij}]$ be a $n \times n$ matrix with $p_{ij} = \omega^{i+j}$. Then $P^2 \neq 0$, when n = **[JEE (Advanced) 2013, Paper-2, (3, –1)/60]**

(A) 57 (B) 55 (C) 58 (D) 56

Paragraph for Question Nos. 10 to 11

Let $S = S_1 \cap S_2 \cap S_3$, where

$S_1 = \{z \in C : |z| < 4\}$, $S_2 = \left\{z \in C : \text{Im}\left[\frac{z-1+\sqrt{3}\,i}{1-\sqrt{3}\,i}\right] > 0\right\}$ and

$S_3 : \{z \in C : \text{Re } z > 0\}$.

10. Area of S = **[JEE (Advanced) 2013, Paper-2, (3, –1)/60]**

(A) $\frac{10\pi}{3}$ (B) $\frac{20\pi}{3}$ (C) $\frac{16\pi}{3}$ (D) $\frac{32\pi}{3}$

11. $\min\limits_{z\in S} |1-3i-z| =$ **[JEE (Advanced) 2013, Paper-2, (3, –1)/60]**

(A) $\frac{2-\sqrt{3}}{2}$ (B) $\frac{2+\sqrt{3}}{2}$ (C) $\frac{3-\sqrt{3}}{2}$ (D) $\frac{3+\sqrt{3}}{2}$

12. Let $z_k = \cos\left(\frac{2k\pi}{10}\right) + i\sin\left(\frac{2k\pi}{10}\right)$; $k = 1, 2,...9$. **[JEE (Advanced) 2014, Paper-2, (3, –1)/60]**

	List I		**List II**
P.	For each z_k there exists a z_j such that $z_k . z_j = 1$	**1.**	True
Q.	There exists a $k \in \{1,2,....,9\}$ such that $z_1 . z = z_k$ has no solution z in the set of complex numbers.	**2.**	False
R.	$\frac{\|1-z_1\|\|1-z_2\|.....\|1-z_9\|}{10}$ equals	**3.**	1
S.	$1 - \sum_{k=1}^{9}\cos\left(\frac{2k\pi}{10}\right)$ equals	**4.**	2

	P	Q	R	S
(A)	1	2	4	3
(B)	2	1	3	4
(C)	1	2	3	4
(D)	2	1	4	3

13. For any integer k, let $\alpha_k = \cos\left(\frac{k\pi}{7}\right) + i\sin\left(\frac{k\pi}{7}\right)$, where $i = \sqrt{-1}$. The value of the expression

$$\frac{\sum_{k=1}^{12}|\alpha_{k+1}-\alpha_k|}{\sum_{k=1}^{3}|\alpha_{4k-1}-\alpha_{4k-2}|}$$ is **[JEE (Advanced) 2015, P-2 (4, 0) / 80]**

14. Let $z = \frac{-1+\sqrt{3}\,i}{2}$, where $i = \sqrt{-1}$ and r, s $\in$ {1,2,3}. Let $P = \begin{bmatrix}(-z)^r & z^{2s}\\ z^{2s} & z^r\end{bmatrix}$ and I be the identity matrix of order 2. Then the total number of ordered pairs (r, s) for which $P^2 = -I$ is
[JEE (Advanced) 2016, Paper-1, (3, 0)/62]

15. Let a, b $\in$ R and $a^2 + b^2 \neq 0$. Suppose $S = \left\{z \in R : z = \frac{1}{a+ibt}, t \in R, t \neq 0\right\}$, where $i = \sqrt{-1}$.
If z = x + iy and z $\in$ S, then (x, y) lies on **[JEE (Advanced) 2016, Paper-2, (4, –2)/62]**
(A) the circle with radius $\frac{1}{2a}$ and centre $\left(\frac{1}{2a}, 0\right)$ for a > 0 , b $\neq$ 0
(B) the circle with radius $-\frac{1}{2a}$ and centre $\left(\frac{1}{2a}, 0\right)$ for a < 0, b $\neq$ 0
(C) the x-axis for a $\neq$ 0, b = 0
(D) the y-axis for a = 0, b $\neq$ 0

16. Let a, b, x and y be real numbers such that a – b = 1 and y $\neq$ 0. If the complex number z = x + *i*y satisfies $\text{Im}\left(\frac{az+b}{z+1}\right) = y$, then which of the following is(are) possible value(s) of x ?
[JEE(Advanced) 2017, Paper-1,(4, –2)/61]
(A) $1-\sqrt{1+y^2}$ (B) $-1-\sqrt{1-y^2}$ (C) $1+\sqrt{1+y^2}$ (D) $-1+\sqrt{1-y^2}$

17. For a non-zero complex number z, let arg(z) denote the principal argument with $-\pi < \arg(z) \leq \pi$. Then, which of the following statement(s) is (are) **FALSE** ?
(A) $\text{Arg}(-1-i) = \frac{\pi}{4}$, where $i = \sqrt{-1}$ **[JEE(Advanced) 2018, Paper-1,(4, –2)/60]**
(B) The function $f : R \to (-\pi, \pi]$, defined by $f(t) = \arg(-1 + it)$ for all t $\in$ R, is continuous at all points of R, where $i = \sqrt{-1}$
(C) For any two non-zero complex numbers z_1 and z_2 , $\arg\left(\frac{z_1}{z_2}\right) - \arg(z_1) + \arg(z_2)$ is an integer multiple of 2π
(D) For any three given distinct complex numbers z_1, z_2 and z_3, the locus of the point z satisfying the condition $\arg\left(\frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}\right) = \pi$, lies on a straight line.

18. Let s, t, r be non-zero complex numbers and L be the set of solutions z = x + iy (x, y $\in$ R, $i = \sqrt{-1}$) of the equation $sz + t\bar{z} + r = 0$, where $\bar{z} = x - iy$. Then,which of the following statement(s) is (are) TRUE ?
(A) If L has exactly one element, then |s| $\neq$ |t|
(B) If |s| = |t|, then L has infinitely many elements **[JEE(Advanced) 2018, Paper-2,(4, –2)/60]**
(C) The number of elements in L $\cap$ {z : |z – 1 + i| = 5} is at most 2
(D) If L has more than one element, then L has infinitely many elements

19. Let S be the set of all complex numbers z satisfying $|z - 2 + i| \geq \sqrt{5}$. If the complex number z_0 is such that $\frac{1}{|z_0 - 1|}$ is the maximum of the set $\left\{\frac{1}{|z-1|} : z \in s\right\}$, then the principal argument of $\frac{4 - z_0 - \bar{z}_0}{z_0 - \bar{z}_0 + 2i}$ is
[JEE(Advanced) 2019, Paper-1,(4, –1)/62]

(A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $-\frac{\pi}{2}$ (D) $\frac{\pi}{2}$

20. That $\omega \neq 1$ be a cube root of unity. Then the minimum of the set $\{|a + b\omega + c\omega^2|^2$; a, b, c are distinct non zero integers$\}$ equals ______. **[JEE(Advanced) 2019, Paper-1,(4, –1)/62]**

PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

1. If α and β are the roots of the equation $x^2 - x + 1 = 0$, then $\alpha^{2009} + \beta^{2009} =$ **[AIEEE 2010, (4, –1), 144]**
(1) – 1 (2) 1 (3) 2 (4) –2

2. The number of complex numbers z such that $|z - 1| = |z + 1| = |z - i|$ equals
[AIEEE 2010, (4, –1), 120]
(1) 1 (2) 2 (3) ∞ (4) 0

3. If $\omega(\neq 1)$ is a cube root of unity, and $(1 + \omega)^7 = A + B\omega$. Then (A, B) equals **[AIEEE 2011, I, (4, –1), 120]**
(1) (0, 1) (2) (1, 1) (3) (1, 0) (4) (–1, 1)

4. Let α, β be real and z be a complex number. If $z^2 + \alpha z + \beta = 0$ has two distinct roots on the line Re z = 1, then it is necessary that : **[AIEEE- 2011, I, (4, –1), 120]**
(1) $\beta \in (0, 1)$ (2) $\beta \in (-1, 0)$ (3) $|\beta| = 1$ (4) $\beta \in (1, \infty)$

5. If z is a complex number of unit modulus and argument θ, then $\arg\left(\frac{1+z}{1+\bar{z}}\right)$ equals :
[AIEEE - 2013, (4, –1), 120]
(1) $-\theta$ (2) $\frac{\pi}{2} - \theta$ (3) θ (4) $\pi - \theta$

6. If z a complex number such that $|z| \geq 2$, then the minimum value of $\left|z + \frac{1}{2}\right|$:
[JEE(Main) 2014,(4,–1), 120]
(1) is strictly greater than 5/2
(2) is strictly greater than 3/2 but less than 5/2
(3) is equal to 5/2
(4) lie in the interval (1, 2)

7. A complex number z is said to be unimodular if $|z| = 1$. Suppose z_1 and z_2 are complex numbers such that $\frac{z_1 - 2z_2}{2 - z_1\bar{z}_2}$ is unimodular and z_2 is not unimodular. Then the point z_1 lies on a :
[JEE(Main) 2015, (4, – 1), 120]
(1) straight line parallel to x-axis
(2) straight line parallel to y-axis
(3) circle of radius 2
(4) circle of radius $\sqrt{2}$

8. A value of θ for which $\frac{2 + 3i \sin\theta}{1 - 2i \sin\theta}$ is purely imaginary, is : **[JEE(Main) 2016, (4, – 1), 120]**
(1) $\frac{\pi}{6}$ (2) $\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (3) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (4) $\frac{\pi}{3}$

9. Let ω be a complex number such that $2\omega + 1 = z$ where $z = \sqrt{-3}$. If $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k$, then k is equal to : **[JEE(Main) 2017, (4, – 1), 120]**
(1) –z (2) z (3) –1 (4) 1

10. If $\alpha, \beta \in C$ are the distinct roots, of the equation $x^2 - x + 1 = 0$, then $\alpha^{101} + \beta^{107}$ is equal to :
[JEE(Main) 2018, (4, – 1), 120]
(1) 1 (2) 2 (3) –1 (4) 0

11. Let α and β be two roots of the equation $x^2 + 2x + 2 = 0$, then $\alpha^{15} + \beta^{15}$ is equal to :
[JEE(Main) 2019, Online (09-01-19),P-1 (4, – 1), 120]
(1) 512 (2) –256 (3) 256 (4) –512

12. Let z be a complex number such that $|z| + z = 3 + i$, (where $i = \sqrt{-1}$) then $|z|$ is equal to :
[JEE(Main) 2019, Online (11-01-19),P-2 (4, – 1), 120]
(1) $\frac{\sqrt{34}}{3}$ (2) 5/4 (3) 5/3 (4) $\frac{\sqrt{41}}{4}$

13. Let z_1 and z_2 be two complex numbers satisfying $|z_1| = 9$ and $|z_2 - 3 - 4i| = 4$. Then the minimum value of $|z_1 - z_2|$ is : **[JEE(Main) 2019, Online (12-01-19),P-2 (4, – 1), 120]**
(1) 0 (2) 1 (3) $\sqrt{2}$ (4) 2

14. If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}\left(i = \sqrt{-1}\right)$, then $(1 + iz + z^5 + iz^8)^9$ is equal to :
[JEE(Main) 2019, Online (08-04-19),P-2 (4, – 1), 120]
(1) 0 (2) $(-1 + 2i)^9$ (3) –1 (4) 1

15. Let $z \in C$ be such that $|z| < 1$, If $\omega = \frac{5+3z}{5(1-z)}$ then :
[JEE(Main) 2019, Online (09-04-19),P-2 (4, – 1), 120]
(1) $4\,\text{Im}(\omega) > 5$ (2) $5\text{Re}(\omega) > 4$ (3) $5\,\text{Re}(\omega) > 1$ (4) $5\,\text{Im}(\omega) < 1$

16. If z and ω are two complex numbers such that $|z\omega| = 1$ and $\arg(z) - \arg(\omega) = \frac{\pi}{2}$, then:
[JEE(Main) 2019, Online (10-04-19),P-2 (4, – 1), 120]
(1) $z\overline{\omega} = -i$ (2) $z\overline{\omega} = \frac{1-i}{\sqrt{2}}$ (3) $z\overline{\omega} = \frac{-1+i}{\sqrt{2}}$ (4) $z\overline{\omega} = i$

17. If the equation, $x^2 + bx + 45 = 0$ $(b \in R)$ has conjugate complex roots and they satisfy $|z + 1| = 2\sqrt{10}$, then : **[JEE(Main) 2020, Online (08-01-20),P-1 (4, –1), 120]**
(1) $b^2 - b = 30$ (2) $b^2 + b = 72$ (3) $b^2 - b = 42$ (4) $b^2 + b = 12$

18. Let $\alpha = \frac{-1+i\sqrt{3}}{2}$ if $a = (1 + \alpha)\sum_{k=0}^{100}\alpha^{2k}$ and $b = \sum_{k=0}^{100}\alpha^{3k}$, then a and b are the roots of the quadratic equation : **[JEE(Main) 2020, Online (08-01-20),P-2 (4, –1), 120]**
(1) $x^2 + 101x + 100 = 0$ (2) $x^2 - 102x + 101 = 0$
(3) $x^2 - 101x + 100 = 0$ (4) $x^2 + 102x + 101 = 0$

19. Let z be a complex number such that $\left|\frac{z-1}{z+2i}\right| = 1$ and $|z| = \frac{5}{2}$. Then the value of $|z + 3i|$ is :
[JEE(Main) 2020, Online (09-01-20),P-1 (4, –1), 120]
(1) $\sqrt{10}$ (2) $2\sqrt{3}$ (3) $\frac{7}{2}$ (4) $\frac{15}{4}$

Answers

EXERCISE - 1
PART - I

Section (A) :

A-1. (i) 3, –1 (ii) x = 1 and y = 2;

(iii) (1, 1) $\left(0, \frac{5}{2}\right)$ (iv) $x = K,\ y = \frac{3K}{2}$, $K \in R$

A-2. (i) 8 (ii) 10π

A-3. (i) [(–2, 2) ; (–2, –2)] (ii) – (77 + 108 i)

A-4. (i) ± (4 + 3i) (ii) $\pm\sqrt{2}$ + 0i or 0 $\pm\sqrt{2}$ i

A-5. z = (2 + i) or (1 – 3i)

A-6. (i) $\frac{21}{5} - \frac{12}{5}$ i (ii) 3 + 4 i (iii) $\frac{1}{2\left(1+3\cos^2\frac{\theta}{2}\right)} + i\,\frac{-\cot\frac{\theta}{2}}{1+3\cos^2\frac{\theta}{2}}$ (iv) 2

A-7. (i) $\pi e^{i\pi}$ (ii) $5e^{i\frac{\pi}{2}}$ (iii) $2e^{-i\frac{5\pi}{6}}$ c (iv) $2e^{-i\frac{4\pi}{5}}$

A-8. (i) $|z| = 2\cos\frac{9\pi}{25}$ Principal Arg z = $\frac{9\pi}{25}$, arg z = $\frac{9\pi}{25}$ + 2kπ, k ∈ I

(ii) Modulus = 2, Arg = $2k\pi - \frac{5\pi}{6}$, k ∈ I, Principal Arg = $-\frac{5\pi}{6}$

(iii) Modulus = $\sec^2 1$, arg = 2kπ + (2 – π), Principal Arg = (2 – π)

(iv) Modulus = $\frac{1}{\sqrt{2}}$ cosec $\frac{\pi}{5}$, arg z = 2kπ + $\frac{11\pi}{20}$, Principal Arg = $\frac{11\pi}{20}$

A-9. $\frac{iz}{2} + \frac{1}{2} + i$ **A-10.** (i) 4 (ii) $\sqrt{3}$

Section (B) :

B-1. (i) $(a - ib)^2$ **B-2.** π **B-3.** 1 **B-7.** 0 **B-8.** (i) $\pm\frac{\pi}{2}$ (ii) 1

B-10. arg z ∈ $\left[-\frac{\pi}{2}, -\frac{\pi}{4}\right]$

Section (C) :

C-1. $\sqrt{5} + 2$ & $\sqrt{5} - 2$

C-2. (i) 0, 6 (ii) 1, 7 (iii) 0, 5

C-3. (i) The region between the concentric circles with centre at (0, 2) & radii 1 and 3 units
(ii) The part of the complex plane on or above the line y = 1
(iii) a ray emanating from the point (3 + 4i) directed away from the origin & having equation, $\sqrt{3}\,x - y + 4 - 3\sqrt{3} = 0,\ x > 3$
(iv) Region outside or on the circle with centre $\frac{1}{2} + 2i$ and radius $\frac{1}{2}$

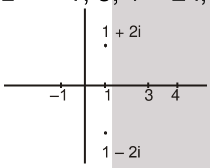
C-4. (i) |z| = 20 (ii) OP = OQ = PR = QR = 20 **C-7.** $9\frac{\pi}{\sqrt{2}}$ **C-8.** 5

C-9. $-4 - 3i,\ 2\sqrt{5}$ **C-10.** a rhombous but not a square

Section (D) :

D-1. 3 **D-2.** –5 **D-3.** 4^n **D-4.** (i) 1 (ii) 1

D-6. z = – 1, 3, 1 – 2 i, 1 + 2 i



Sum = 4
centroid = 1

D-7. (i) – 1 (ii) $e^{(6n+1)\frac{\pi}{4}i}$, n = 0, 1, 2, 3. Continued product = 1

D-8. ω **D-9.** $x^2 + x + 2 = 0$

PART - II

Section (A) :

A-1. (A) **A-2.** (D) **A.3.** (B) **A-4.** (A) **A-5.** (D) **A-6.** (D) **A-7.** (D)
A-8. (B) **A-9.** (A) **A-10.** (A) **A-11.** (C) **A-12.** (D)

Section (B) :

B-1. (A) **B-2.** (C) **B-3.** (A) **B-4.** (D) **B-5.** (D) **B-6.** (A) **B-7.** (D)
B-8. (D) **B-9.** (C) **B-10.** (C) **B-11.** (A) **B-12.** (B) **B-13.** (D)

Section (C) :

C-1. (A) **C-2.** (D) **C-3.** (D) **C-4.** (A) **C-5.** (B) **C-6.** (A) **C-7.** (C)
C-8. (B) **C-9.** (C) **C-10.** (A) **C-11.** (B) **C-12.** (B) **C-13.** (C)

Section (D) :

D-1. (B) **D-2.** (C) **D-3.** (A) **D-4.** (A) **D-5.** (A) **D-6.** (C) **D-7.** (A)
D-8. (C)

PART - III

1. (A) → (s), (B) → (r), (C) → (q), (D) → (p) **2.** A → s; B → r; C → p; D → q.
3. a → p, r; b → p,q,r, t; c → p,r,s; d → p,q,r,s, t. **4.** (A) → (p),(B) → (q), (C) → (r), (D) → (s)

EXERCISE - 2

PART - I

1.	(B)	**2.**	(A)	**3.**	(A)	**4.**	(D)	**5.**	(B)	**6.**	(B)	**7.**	(C)
8.	(C)	**9.**	(A)	**10.**	(A)	**11.**	(B)	**12.**	(B)	**13.**	(A)	**14.**	(C)
15.	(C)	**16.**	(B)	**17.**	(D)								
18.	(B)												

PART - II

1.	39.60	2.	02.35 or 02.36	3.	13.00	4.	25.00	5.	09.11	6.	01.41
7.	10.00	8.	00.00	9.	02.00	10.	15.00	11.	00.00	12.	00.27
13.	16.00	14.	20.94	15.	16.00	16.	16.00	17.	03.28 or 03.29		
18.	01.00	19.	02.93	20.	07.00						

PART - III

1.	(ABC)	**2.**	(ABC)	**3.**	(BD)	**4.**	(BC)	**5.**	(BD)	**6.**	(BCD)
7.	(ABC)	**8.**	(ABCD)	**9.**	(A BD)	**10.**	(ABCD)	**11.**	(ABCD)	**12.**	(AB)
13.	(BCD)	**14.**	(ABC)	**15.**	(ABCD)	**16.**	(ABC)	**17.**	(AB)	**18.**	(ACD)
19.	(ABCD)	**20.**	(ACD)	**21.**	(ABC)	**22.**	(ABC)	**23.**	(ABCD)	**24.**	(CD)
25.	(BC)	**26.**	(ABC)	**27.**	(ABCD)						

PART - IV

1.	(B)	**2.**	(D)	**3.**	(A)	**4.**	(D)	**5.**	(C)	**6.**	(D)
7.	(A)	**8.**	(B)	**9.**	(A)	**10.**	(B)				

EXERCISE - 3

PART - I

1*. (ACD) **2.** 1

3. (A) - (q,r), (B)-(p), (C) - (p,s,t), (D) - (q,r,s,t) **4.** **(5)**

5. Bonus (w = $e^{i\pi/3}$ is a typographical error, because of this the answer cannot be an integer.)
(if w = then answer comes out to be 3)

6.	(D)	**7.**	(C)	**8.**	(C)	**9.**	(BCD)	**10.**	(B)
11.	(C)	**12.**	(C)	**13.**	4	**14.**	1	**15.**	(A,C,D)
16.	(B,D)	**17.**	(ABD)	**18.**	(ACD)	**19.**	(C)	**20.**	(3)

PART - II

1.	(2)	**2.**	(1)	**3.**	(2)	**4.**	(4)	**5.**	(3)	**6.**	(4)	**7.**	(3)
8.	(3)	**9.**	(1)	**10.**	(1)	**11.**	(2)	**12.**	(3)	**13.**	(1)	**14.**	(3)
15.	(3)	**16.**	(1)	**17.**	(1)	**18.**	(2)	**19.**	(3)				